

Notes on mixed autonomous/human traffic in the Nagel–Schreckenberg model

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1 Purpose of this note

This note collects the derivations behind the study of mixed autonomous-vehicle (AV) and human-driven-vehicle (HV) traffic in a Nagel–Schreckenberg (NaSch) cellular automaton. The manuscript reports the results; here the intermediate steps are written out so that every number can be traced back to an equation or to a stored simulation. Three topics are treated: (i) the behavior of four AV car-following rules under a common safety floor, (ii) the exact solution of the mixed model at $v_{\max} = 1$, and (iii) the slow-to-start hysteresis and its suppression by AV penetration. In section 6, a simple jam-outflow argument is developed which reproduces the jammed branch of the simulations without fitted parameters (rms deviation about 0.0015 in flux) and explains the shape of the hysteresis suppression curve in closed form.

Throughout, the lattice spacing is one cell (7.5 m) and the time step is $\Delta t = 1$ (1 s). The AV fraction (penetration rate) is denoted by r , and the density by $\rho = N/L$ for N vehicles on a ring of L cells.

2 Nagel–Schreckenberg model

2.1 Update rules

Vehicle i has position x_i and integer velocity $v_i \in \{0, \dots, v_{\max}\}$. Let

$$d_i \equiv x_{i+1} - x_i \pmod{L} \quad (1)$$

be the center-to-center distance to the leader $i + 1$, so that $d_i - 1$ is the number of empty cells in between. One time step consists of

$$v_i \leftarrow \min(v_i + 1, v_{\max}), \quad (2)$$

$$v_i \leftarrow \min(v_i, d_i - g_{\min}), \quad (3)$$

$$v_i \leftarrow \max(v_i - 1, 0) \text{ with probability } p, \quad (4)$$

$$x_i \leftarrow x_i + v_i, \quad (5)$$

applied to all vehicles in parallel. The original model [Nagel and Schreckenberg, 1992] has $g_{\min} = 1$. In this study $g_{\min} = 1$ is used for every vehicle type; the reason is given in section 3.2. An HV has $v_{\max} = 5$ and $p = 0.5$ unless stated otherwise.

2.2 Deterministic limit

For $p = 0$ the stationary flux is obtained exactly. If all gaps satisfy $d_i - 1 \geq v_{\max}$, every vehicle moves with $v_{\max}$ and the configuration simply translates. This is possible only if $L/N \geq v_{\max} + 1$, namely $\rho \leq \rho_c \equiv 1/(v_{\max} + 1)$, and the flux is $J = \rho v_{\max}$. For $\rho > \rho_c$ the stationary state has $v_i = d_i - 1$ for every vehicle, i.e., every vehicle advances exactly by its own gap. The total displacement per step is then $\sum_i (d_i - 1) = L - N$, and the flux per site is

$$J = \frac{L - N}{L} = 1 - \rho. \quad (6)$$

Combining the two regimes,

$$J_{\det}(\rho) = \min(v_{\max}\rho, 1 - \rho), \quad J_{\det, \max} = \frac{v_{\max}}{v_{\max} + 1} \quad \text{at} \quad \rho_c = \frac{1}{v_{\max} + 1}. \quad (7)$$

For $v_{\max} = 5$, $J_{\det, \max} = 5/6 = 0.833$ at $\rho_c = 0.167$. The simulated all-AV capacity under the naive deterministic rule is 0.830 (table 1), in agreement with (7) up to the density grid. For $p = 0.5$ no closed form exists at $v_{\max} = 5$; the simulation gives $J_{\max} = 0.328$ at $\rho = 0.075$, close to the value 0.32 quoted by Nagel and Schreckenberg [1992].

3 Mixed AV/HV traffic

3.1 Assignment of vehicle types

Out of $N = \text{round}(\rho L)$ vehicles, $N_{\text{AV}} = \text{round}(rN)$ are chosen uniformly at random as AVs. Since vehicles cannot pass each other, the cyclic order of AVs and HVs is fixed during a run. A new random order is drawn for each seed, so that the statistical errors include the effect of the sequence. In the simplest (naive) AV model, an AV obeys (2)–(5) with $p = 0$; it differs from an HV only by the absence of random slow-down.

3.2 Collision safety under parallel update

A frequently used way of representing an AV is a smaller safety margin, $g_{\min, \text{AV}} < g_{\min, \text{HV}}$. This is unsafe under the parallel update. Consider a follower f and a leader l at distance $d = x_l - x_f$ before the update. Rule (3) gives $v_f \leq d - g_{\min, f}$, evaluated against the *pre-update* position of the leader. After the update, the distance is

$$d' = d - v_f + v_l \geq g_{\min, f} + v_l. \quad (8)$$

The two vehicles do not overlap if and only if $d' \geq 1$. Since the leader may realize $v_l = 0$ (by random slow-down, or by its own gap constraint), the condition $d' \geq 1$ is guaranteed for every possible $v_l \geq 0$ only if $g_{\min, f} \geq 1$. With $g_{\min, \text{AV}} = 0$, a collision occurs whenever an AV closes to zero gap behind a leader that does not move.

In a homogeneous AV population with $p = 0$ the leader of a closing AV is itself deterministic and the failure does not show up. In a mixed population it is frequent: for $\rho = 0.1$ and $r = 0.2$, duplicate positions were found in 2515 of 3000 measured steps. Hence $g_{\min} = 1$ is used for all vehicles, and the AV/HV difference is expressed only through the car-following dynamics.

3.3 Common average velocity on a ring

Since $1 \leq x_{i+1}(t) - x_i(t) \leq L$ for all t (positions unwrapped), we have

$$\bar{v}_{i+1} - \bar{v}_i = \lim_{T \rightarrow \infty} \frac{[x_{i+1}(T) - x_i(T)] - [x_{i+1}(0) - x_i(0)]}{T} = 0, \quad (9)$$

where $\bar{v}_i$ is the long-time average velocity of vehicle i . Hence all vehicles share one average velocity $\bar{v} = J/\rho$, regardless of type. This fact is exact and is used repeatedly below: in section 5 it closes the two-species gap equations, and in section 6 it justifies the treatment of the jam front.

4 Car-following rules on the lattice

4.1 Shared safety floor and net spacing

Each rule computes a desired velocity $v_{i,\text{des}}$ and is then clipped by the common floor

$$v_i \leftarrow \min(v_{i,\text{des}}, s_i), \quad s_i \equiv d_i - 1, \quad (10)$$

where s_i is the net (bumper-to-bumper) spacing. AVs use $p = 0$ in (4). IDM, ACC, and CACC are formulated in the net spacing in their original papers, so s_i (and not d_i) must be passed to them. Supplying d_i gives each rule one extra cell, 7.5 m, of apparent room.

A useful way to read (10) is in terms of time headway. The floor allows $v_i = s_i$, i.e., a vehicle may cover its whole gap in one step. The floor therefore corresponds to a minimum time headway of exactly one time step (1 s). A control law that asks for a time headway shorter than one step is clipped by the floor; one that asks for a longer headway is not. This observation explains the difference between ACC and CACC below.

The velocity of the continuous rules is kept as a floating-point state, and only the displacement is rounded to an integer number of cells. The flux is measured from the integer displacement.

4.2 Unit conversion

With 1 cell = 7.5 m and 1 step = 1 s,

$$v_{\text{max}} = 5 \text{ cell/step} = 37.5 \text{ m/s}, \quad a = \frac{0.73}{7.5} = 0.097, \quad b = \frac{1.67}{7.5} = 0.223 \text{ cell/step}^2, \quad T = 1.6 \text{ step}. \quad (11)$$

The ACC/CACC gains and time headways h carry over unchanged, since one step is one second.

4.3 Intelligent Driver Model

The Euler-discretized IDM [Treiber et al., 2000; Helbing, 2001] reads

$$v_{i,\text{des}} = v_i + a \left[1 - \left(\frac{v_i}{v_0} \right)^\delta - \left(\frac{s^*(v_i, \Delta v_i)}{s_i} \right)^2 \right], \quad s^* = s_0 + \max \left(0, T v_i + \frac{v_i \Delta v_i}{2\sqrt{ab}} \right), \quad (12)$$

with $\Delta v_i = v_i - v_{i+1}$, $v_0 = v_{\text{max}}$, $s_0 = 0$, and $\delta = 4$. In a uniform stationary state ($v_{i,\text{des}} = v_i$, $\Delta v_i = 0$) the bracket vanishes, which gives the equilibrium spacing

$$s_e(v) = \frac{Tv}{\sqrt{1 - (v/v_0)^4}}. \quad (13)$$

Since $s_e \rightarrow \infty$ as $v \rightarrow v_0$, an IDM vehicle never reaches $v_{\max}$ at finite spacing. With $s = 1/\rho - 1$ the equilibrium flux is $J = \rho v = v/(1 + s_e(v))$. Maximizing numerically over v ,

$$J_{\text{IDM,max}}^{\text{eq}} = 0.490 \quad \text{at} \quad v = 2.79, \quad \rho = 0.176. \quad (14)$$

The simulated all-IDM capacity is 0.422, i.e., the uniform equilibrium is not fully realized on the lattice.

Rounding gridlock. If the equilibrium velocity drops below $1/2$, the rounded displacement is zero, nothing moves, the spacings stop changing, and the state is absorbing. From (13), $s_e(1/2) = 0.80$, which corresponds to $\rho = 1/(1 + 0.80) = 0.56$ for a uniform state. In the simulation the IDM flux vanishes identically for $\rho \gtrsim 0.65$ – 0.7 ; the uniform estimate gives the same order of magnitude. Carrying the fractional remainder of the displacement to the next step removes the gridlock (the IDM flux at $\rho = 0.70$ becomes 0.074) and changes the capacities only slightly, since they are attained at $\rho \approx 0.07$ – 0.17 .

4.4 ACC

The ACC law of Milanés and Shladover [2014] is

$$v_{i,\text{des}} = v_i + k_1 e_i + k_2 (v_{i+1} - v_i), \quad e_i = s_i - h v_i, \quad k_1 = 0.23, \quad k_2 = 0.07, \quad h = 1.1. \quad (15)$$

At a uniform fixed point ($v_{i+1} = v_i$, $v_{i,\text{des}} = v_i$) we need $e_i = 0$, so that

$$v_e = \frac{s}{h} = \frac{s}{1.1} < s. \quad (16)$$

The equilibrium lies below the floor, and ACC is *not* floor-bound. Its equilibrium fundamental diagram is

$$J_{\text{ACC}}^{\text{eq}} = \min\left(v_{\max}\rho, \frac{1-\rho}{h}\right), \quad J_{\text{ACC,max}}^{\text{eq}} = \frac{v_{\max}}{1 + h v_{\max}} = 0.769 \quad \text{at} \quad \rho = \frac{1}{1 + h v_{\max}} = 0.154. \quad (17)$$

The simulated all-ACC capacity is 0.478, far below this value. Since the displacement in (5) uses the updated velocity, a single ACC vehicle behind a leader of constant speed obeys, for small deviations $(\delta v_i, \delta s_i)$, the linear map with Jacobian

$$\begin{pmatrix} A & k_1 \\ -A & 1 - k_1 \end{pmatrix}, \quad A \equiv 1 - k_1 h - k_2 = 0.677. \quad (18)$$

Its eigenvalues are complex with modulus $\sqrt{A} = 0.82$, so the response of a single vehicle is stable. The gap between 0.769 and 0.478 must therefore come from collective effects (platoon instability) or from displacement rounding. It has not been examined and is left as future work.

If the gross distance $d_i = s_i + 1$ is supplied instead, the fixed point becomes $v_e = (s + 1)/h$, which exceeds the floor s whenever $s < 1/(h - 1) = 10$. The gross convention therefore pushes ACC toward the floor-bound regime at all relevant densities, which is consistent with its higher simulated capacity under that convention (0.611 at $r = 1$).

4.5 CACC

The CACC law of Milanés and Shladover [2014] is

$$v_{i,\text{des}} = v_i + k_p e_i + k_d (e_i - e_i^{\text{prev}}), \quad e_i = s_i - h v_i, \quad k_p = 0.45, \quad k_d = 0.25, \quad h = 0.6. \quad (19)$$

At a fixed point $e_i = e_i^{\text{prev}} = 0$, so again $v_e = s/h$. Now $h < 1$, and

$$v_e = \frac{s}{0.6} > s \quad \text{for every } s > 0. \quad (20)$$

The control law always requests more than the floor permits, and the stationary velocity is set by the floor, $v = \min(v_{\text{max}}, s)$. This is exactly the stationary state of the naive deterministic rule (section 2.2). Hence the CACC fundamental diagram coincides with (7), and $J_{\text{max}} = 5/6$. The simulation gives 0.830 for both rules at $r = 1$. The coincidence is partly helped by rounding: with the fractional-carry displacement CACC gives 0.763 at $r = 1$. The ordering $\text{CACC} \approx \text{naive} \gg \text{ACC}$ is robust under both schemes.

In terms of the time-headway reading of the floor: CACC requests $h = 0.6 \text{ s} < 1 \text{ s}$ and is clipped; ACC requests $h = 1.1 \text{ s} > 1 \text{ s}$ and is not. The distinguishing feature of CACC, its short headway, is precisely the part that the lattice cannot represent.

4.6 Simulated capacities

Table 1 and figure 1 give $J_{\text{max}}(r)$ for the four rules (net spacing, $L = 1000$, 16 seeds). The naive and CACC rules raise the capacity monotonically and convexly up to $5/6$. ACC raises it less. IDM with the human-calibrated a , b is non-monotonic: the mixed population carries less flux than either pure population. This deficit disappears when a and b are doubled (table 2), so it is a calibration effect rather than a fixed property of IDM.

Table 1: Capacity $J_{\text{max}}(r)$ by AV rule. Standard errors over 16 seeds in parentheses.

r	naive	IDM	ACC	CACC
0.00	0.327 (0.002)	0.334 (0.003)	0.330 (0.002)	0.333 (0.002)
0.25	0.388 (0.002)	0.276 (0.004)	0.361 (0.003)	0.390 (0.003)
0.50	0.473 (0.002)	0.289 (0.004)	0.404 (0.003)	0.476 (0.002)
0.75	0.604 (0.003)	0.339 (0.006)	0.448 (0.000)	0.606 (0.002)
1.00	0.830 (0.000)	0.422 (0.000)	0.478 (0.006)	0.830 (0.000)
equilibrium estimate ($r = 1$)	$5/6 = 0.833$	0.490	0.769	$5/6 = 0.833$

Table 2: IDM calibration sensitivity of $J_{\text{max}}(r)$ (independent seed stream).

calibration	a (m/s ²)	b (m/s ²)	T (s)	$r = 0$	$r = 0.5$	$r = 1$
baseline	0.73	1.67	1.6	0.328	0.288	0.424
responsive	1.46	3.34	1.6	0.335	0.344	0.444
short headway	0.73	1.67	0.8	0.329	0.278	0.291
combined	1.46	3.34	0.8	0.332	0.346	0.475

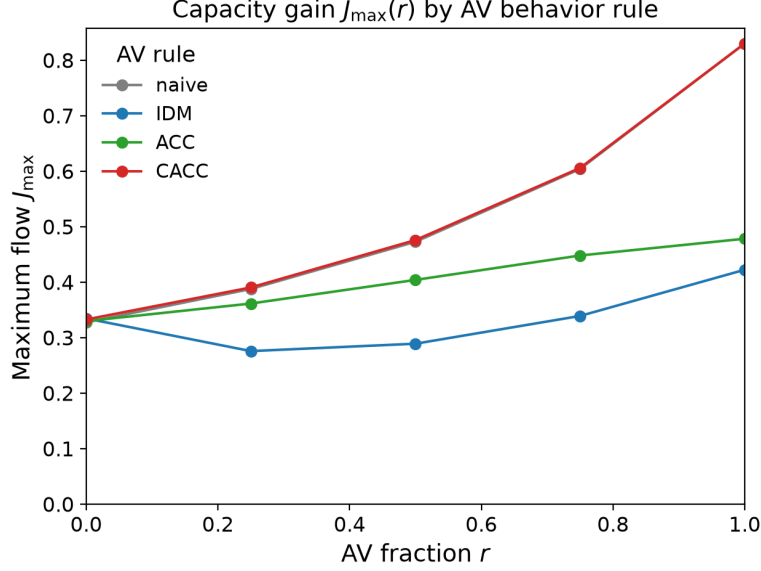


Figure 1: Capacity $J_{\max}(r)$ for the four AV rules ($v_{\max} = 5$, $p_{\text{HV}} = 0.5$).

5 Exact solution at $v_{\max} = 1$

At $v_{\max} = 1$ the mixed model is a totally asymmetric exclusion process on a ring with particle-wise disordered hopping probabilities and parallel update. This class was solved exactly by Evans [1997] for an arbitrary distribution of hopping probabilities; the two-species case below is a special case of his result. Here it is rederived in the car-oriented mean-field (COMF) language of Schadschneider and Schreckenberg [1997], and a closed form for the two-species flux is obtained.

5.1 Gap master equation

At $v_{\max} = 1$, rules (2)–(4) reduce to the following: a vehicle with at least one empty cell ahead moves with probability $q = 1 - p$, and a vehicle with no empty cell ahead does not move. Let P_n be the stationary probability that a vehicle has n empty cells ahead, and let g be the probability that its leader moves in a given step ($\bar{g} \equiv 1 - g$). In one step the gap changes as $n \rightarrow n + (\text{leader moved}) - (\text{self moved})$. Treating the leader's move as independent of the follower's gap (the COMF assumption, which is exact for this model [Schadschneider and Schreckenberg, 1997; Evans, 1997]), the stationary balance reads

$$P_0 = \bar{g} P_0 + \bar{g} q P_1, \quad (21)$$

$$P_1 = g P_0 + (gq + \bar{g}p) P_1 + \bar{g} q P_2, \quad (22)$$

$$P_n = gp P_{n-1} + (gq + \bar{g}p) P_n + \bar{g} q P_{n+1}, \quad n \geq 2. \quad (23)$$

In (21), a zero gap persists if the leader stays (the follower cannot move anyway), and a unit gap closes if the leader stays while the follower moves. In (22), the transition $0 \rightarrow 1$ has weight g rather than gp , since a vehicle with zero gap cannot move.

5.2 Solution

From (21),

$$P_1 = \frac{g}{\bar{g}q} P_0. \quad (24)$$

For $n \geq 2$ we try $P_n = P_1 z^{n-1}$. Substituting into (23) gives

$$\bar{g}q z^2 - (1 - gq - \bar{g}p)z + gp = 0. \quad (25)$$

Since $\bar{g}q - (1 - gq - \bar{g}p) + gp = p + q - 1 = 0$, one root is $z = 1$, which is not normalizable. The other root is

$$z = \frac{gp}{\bar{g}q}. \quad (26)$$

Equation (22) must then be consistent with (24). Using $\bar{g}qz = gp$,

$$P_1 (1 - gq - \bar{g}p - gp) = gP_0 \implies P_1 \bar{g}q = gP_0, \quad (27)$$

which is identical to (24). Hence

$$P_n = \frac{g}{\bar{g}q} z^{n-1} P_0 \quad (n \geq 1). \quad (28)$$

Normalization, $P_0[1 + g/(\bar{g}q(1 - z))] = 1$ with $\bar{g}q(1 - z) = \bar{g}q - gp$, gives

$$P_0 = \frac{\bar{g}q - gp}{\bar{g}q - gp + g} = \frac{\bar{g}q - gp}{q} = \bar{g} - \frac{p}{q} g. \quad (29)$$

The vehicle's own move probability is then

$$q(1 - P_0) = q - q\bar{g} + pg = g. \quad (30)$$

Equation (30) states that a vehicle following a leader that moves with probability g itself moves with probability g : self-consistency is built into the solution and does not fix g . The value of g is fixed by the density.

5.3 Mean headway

Every vehicle occupies its own cell plus the empty cells ahead, so $1/\rho = \langle n + 1 \rangle = 1 + \langle n \rangle$. From (28),

$$\langle n \rangle = \frac{gP_0}{\bar{g}q} \sum_{n \geq 1} n z^{n-1} = \frac{gP_0}{\bar{g}q(1 - z)^2}. \quad (31)$$

With $P_0 = (\bar{g}q - gp)/q$ and $1 - z = (\bar{g}q - gp)/(\bar{g}q)$,

$$\langle n \rangle = \frac{g\bar{g}}{\bar{g}q - gp} = \frac{g(1 - g)}{q - g}. \quad (32)$$

For a deterministic vehicle ($q = 1$), (32) gives $\langle n \rangle = g$, with $P_0 = 1 - g$, $P_1 = g$, and $P_n = 0$ for $n \geq 2$: an AV never lets its gap exceed one cell. Requiring $P_0 \geq 0$ in (29) gives $g \leq q$: a vehicle cannot move more often than its own q .

5.4 Single species

For one species, $1/\rho - 1 = g(1 - g)/(q - g)$, i.e.,

$$\rho g^2 - g + (1 - \rho)q = 0. \quad (33)$$

The root with $g \leq q$ gives $J = \rho g$,

$$J(\rho) = \frac{1}{2} \left[1 - \sqrt{1 - 4q\rho(1 - \rho)} \right], \quad (34)$$

which is the known exact result [Schadschneider and Schreckenberg, 1997]. For $p = 0.5$ the maximum is $J_{\max} = (1 - \sqrt{1 - q})/2 = 0.146$ at $\rho = 1/2$.

5.5 Two species

Now let a fraction r of vehicles have $q_{\text{AV}} = 1$ and a fraction $1 - r$ have $q_{\text{HV}} = q < 1$. By section 3.3, every vehicle moves with the same long-time probability g , so the leader of any vehicle moves with probability g irrespective of its type. Each type therefore satisfies (21)–(23) with its own q_τ and the common g , and (30) holds for each type separately. The density constraint is

$$\frac{1}{\rho} = 1 + r \langle n \rangle_{\text{AV}} + (1 - r) \langle n \rangle_{\text{HV}} = 1 + rg + (1 - r) \frac{g(1 - g)}{q - g}, \quad (35)$$

with the physical range $0 < g < q$. Multiplying by $\rho(q - g)$ and collecting terms,

$$\rho g^2 - (1 - r\rho)g + (1 - \rho)q = 0, \quad (36)$$

so that the flux is obtained in closed form,

$$J(\rho, r) = \frac{1}{2} \left[1 - r\rho - \sqrt{(1 - r\rho)^2 - 4q\rho(1 - \rho)} \right], \quad 0 \leq r < 1. \quad (37)$$

At $r = 0$, (37) reduces to (34). Equation (37) agrees with the numerical root of (35) used in the manuscript to 2×10^{-13} . For a general distribution of hopping probabilities, (35) generalizes to

$$\frac{1}{\rho} - 1 = g(1 - g) \left\langle \frac{1}{q_\mu - g} \right\rangle_\mu, \quad (38)$$

which is the self-consistency condition of Evans [1997] in COMF form.

5.6 The limit $r \rightarrow 1$

At $r = 1$ there are no HVs, the HV term in (35) is absent, and $1/\rho = 1 + g$ gives $J = \min(\rho, 1 - \rho)$, i.e., (7) with $v_{\max} = 1$. The limit $r \rightarrow 1^-$ of (37) is different. Setting $r = 1$ in the quadratic, its two roots are $g = q$ and $g = (1 - \rho)/\rho$, and the physical root is the smaller one,

$$\lim_{r \rightarrow 1^-} J(\rho, r) = \min(q\rho, 1 - \rho), \quad \lim_{r \rightarrow 1^-} J_{\max} = \frac{q}{1 + q} \quad \text{at} \quad \rho = \frac{1}{1 + q}. \quad (39)$$

For $p = 0.5$ this is $1/3$ at $\rho = 2/3$, against $1/2$ at $\rho = 1/2$ for $r = 1$. The capacity is therefore discontinuous at $r = 1$ (figure 2b). The reason is the bound $g < q_{\text{HV}}$: in a ring with a finite fraction of HVs, no vehicle can move more often than the HVs do, and in free flow all vehicles queue up

behind HVs and move with the HV speed q . Since $\langle 1/(q_\mu - g) \rangle$ diverges as $g \rightarrow q_{\text{HV}}$ for any finite HV fraction, every density is reachable with $g < q_{\text{HV}}$, and no condensation transition of the type discussed by Evans [1997] occurs. In a finite ring, the discontinuity is smoothed into a crossover when only a few HVs remain.

The particle-hole symmetry $J(\rho) = J(1 - \rho)$ of the single-species model is broken for $0 < r < 1$, and the density of maximum flux moves above $1/2$ (table 3).

Table 3: Maximum flux of (37) at $v_{\text{max}} = 1$, $p_{\text{HV}} = 0.5$.

r	0	0.2	0.4	0.6	0.8	0.9	0.99	1
J_{max}	0.146	0.158	0.173	0.193	0.224	0.250	0.304	0.500
ρ at max	0.500	0.516	0.535	0.558	0.590	0.613	0.650	0.500

5.7 Comparison with simulation

Figure 2a compares (37) with simulations at $v_{\text{max}} = 1$, $p_{\text{HV}} = 0.5$, $L = 2000$, for $r = 0, 0.2, \dots, 1$. The maximum relative error is below 0.53% at every r , and the mean relative error is below 0.2%. Since the simulation uses the same safety floor (10) and naive AV rule as the $v_{\text{max}} = 5$ runs, this comparison verifies the simulation framework in the one regime where an exact result is available.

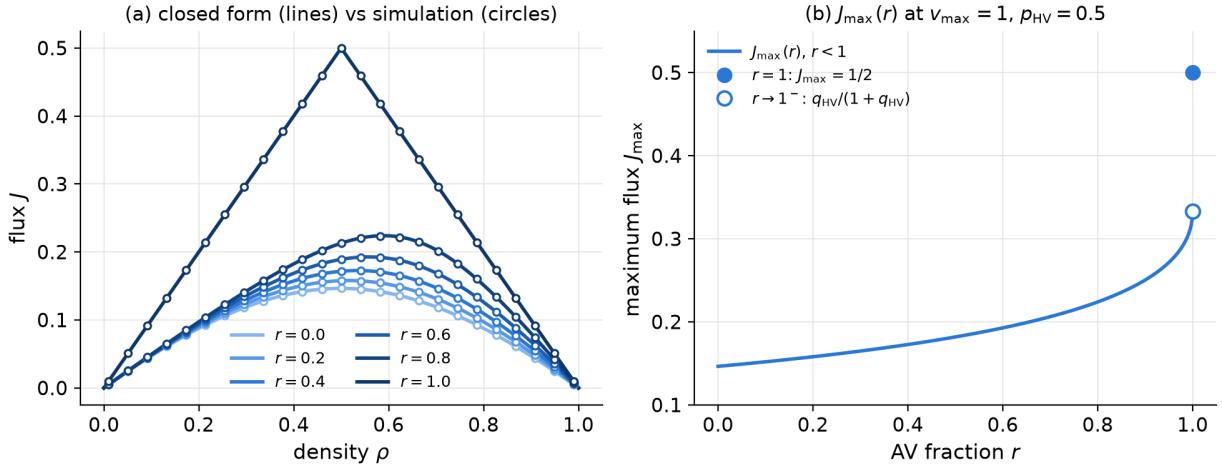


Figure 2: (a) Two-species flux (37) against simulation at $v_{\text{max}} = 1$, $p_{\text{HV}} = 0.5$. (b) Maximum flux as a function of r . The filled circle is $r = 1$ and the open circle is the limit $r \rightarrow 1^-$.

6 Slow-to-start rule and hysteresis

6.1 VDR rule

In the velocity-dependent-randomization (VDR) model of Barlović et al. [1998], the slow-down probability in (4) depends on the velocity at the beginning of the step,

$$p(v) = \begin{cases} p_0, & v = 0, \\ p, & v > 0, \end{cases} \quad p_0 = 0.75, \quad p = 1/64, \quad (40)$$

for HVs, while AVs have $p = p_0 = 0$. A standing HV restarts with probability $q_0 \equiv 1 - p_0 = 0.25$ per step, whereas a moving HV rarely brakes. As a consequence, two stationary states coexist in a window of densities: a homogeneous free-flow state (upper branch, reached from a homogeneous initial condition) and a phase-separated state containing a jam (lower branch, reached from a jammed initial condition) [Barlović et al., 1998]. The loop width is defined as

$$W \equiv \max_{\rho} [J_{\text{hom}}(\rho) - J_{\text{jam}}(\rho)]. \quad (41)$$

6.2 Upper branch

In free flow every vehicle moves with nearly v_{max} ; the average free velocity is $v_f \simeq v_{\text{max}} - p$ for an HV (and v_{max} for an AV; the difference is below 0.016 and is neglected). As in section 2.2, a homogeneous state with all gaps $\geq v_{\text{max}}$ exists only for

$$\rho \leq \rho_2 \equiv \frac{1}{v_{\text{max}} + 1} = 0.167, \quad (42)$$

so that $J_{\text{up}}(\rho) = v_f \rho$ for $\rho \leq \rho_2$. In practice, rare braking events nucleate jams before ρ_2 is reached, and the upper branch ends somewhat earlier in a finite measurement time.

6.3 Jam outflow

Consider a compact jam (every cell occupied) with a free road downstream. The vehicle at the head of the jam has $v = 0$ and at least one empty cell ahead. After (2) and (3) its velocity is 1, and by (40) it stays at rest with probability p_0 . It therefore leaves with probability q_0 per step if it is an HV, and with probability 1 if it is an AV. Under the parallel update the next vehicle sees an empty cell from the following step on, so the waiting times of successive head vehicles are independent and geometric. The mean interval between departures is

$$\bar{\tau}(r) = r \cdot 1 + (1 - r) \frac{1}{q_0}, \quad (43)$$

where the type of the head vehicle is AV with probability r (the sequence is random; this is also confirmed below by the measured composition of standing vehicles). The departure rate is

$$c(r) = \frac{1}{\bar{\tau}(r)} = \frac{q_0}{1 - p_0 r}. \quad (44)$$

Each departure moves the jam front upstream by one cell, so the front recedes with speed c . Downstream of the front, vehicles travel with v_f at density ρ_{out} . The number of vehicles crossing the moving front per step must be the same on both sides,

$$c \cdot 1 = \rho_{\text{out}} (v_f + c), \quad (45)$$

which gives

$$\rho_{\text{out}} = \frac{c}{v_f + c}, \quad J_{\text{out}} = \rho_{\text{out}} v_f = \frac{c v_f}{v_f + c}. \quad (46)$$

As a check, the deterministic model ($p_0 = p = 0$, $c = 1$, $v_f = v_{\text{max}}$) gives $J_{\text{out}} = v_{\text{max}}/(v_{\text{max}} + 1) = 5/6$ and $\rho_{\text{out}} = 1/6$, which coincide with the maximum of (7). In the deterministic model the outflow from a jam equals the capacity, and no hysteresis is possible. For the VDR HVs ($r = 0$, $q_0 = 0.25$) we obtain $J_{\text{out}} = 0.238$ and $\rho_{\text{out}} = 0.048$, far below the maximum free flux $v_f \rho_2 \simeq 0.83$.

6.4 Lower branch

For $\rho > \rho_{\text{out}}$, the lower branch is a mixture of a jam (density 1, flux 0) occupying a fraction ϕ of the ring and an outflow region (density ρ_{out} , flux J_{out}) occupying the rest. The lever rule gives

$$\rho = \phi + (1 - \phi)\rho_{\text{out}}, \quad J = (1 - \phi)J_{\text{out}} \implies J_{\text{low}}(\rho) = J_{\text{out}} \frac{1 - \rho}{1 - \rho_{\text{out}}}. \quad (47)$$

From (46), $1 - \rho_{\text{out}} = v_f/(v_f + c)$ and $J_{\text{out}}/(1 - \rho_{\text{out}}) = c$, so the lower branch takes the simple form

$$J_{\text{low}}(\rho, r) = c(r)(1 - \rho) = \frac{q_0}{1 - p_0 r}(1 - \rho), \quad \rho > \rho_{\text{out}}(r). \quad (48)$$

The jammed branch is a straight line through $(\rho, J) = (1, 0)$ whose slope is the upstream speed of the jam front, i.e., the departure rate of the head vehicle. For $\rho < \rho_{\text{out}}$ the jam dissolves and $J_{\text{low}} = v_f \rho$.

Figure 3a compares (48) with the jammed-start simulations ($L = 8000$, 20 seeds) without any fitted parameter. Over the whole scan ($0 \leq r \leq 1$, $0.04 \leq \rho \leq 0.40$, four ring sizes), the rms deviation is about 0.0015 in flux, and the maximum deviation is 0.007 once a thin layer $\rho_{\text{out}} < \rho < \rho_{\text{out}} + 0.03$ is excluded. Inside this layer the jam contains only a few vehicles and can dissolve during the run, so the measured flux lies above (48). This is a finite-size effect: the largest deviation in the layer is 0.10 for $L = 1000$, 0.057 for $L = 2000$, 0.029 for $L = 4000$, and 0.012 for $L = 8000$. Table 4 lists c , J_{out} , and ρ_{out} .

6.5 Loop width

The two branches coexist for $\rho_{\text{out}}(r) < \rho < \rho_2$, and the largest separation is at ρ_2 . The ideal loop width is

$$W_{\text{id}}(r) = v_f \rho_2 - c(r)(1 - \rho_2) \simeq \frac{v_{\text{max}}}{v_{\text{max}} + 1} [1 - c(r)] = \frac{v_{\text{max}}}{v_{\text{max}} + 1} \frac{p_0(1 - r)}{1 - p_0 r}, \quad (49)$$

where $v_f \simeq v_{\text{max}}$ was used in the second step. For $p_0 = 0.75$, (49) gives $W_{\text{id}}(0) = 0.625$ and $W_{\text{id}}(1) = 0$. Its derivative is

$$\frac{dW_{\text{id}}}{dr} = -\frac{v_{\text{max}}}{v_{\text{max}} + 1} \frac{p_0(1 - p_0)}{(1 - p_0 r)^2}, \quad (50)$$

which is finite on the whole interval $0 \leq r \leq 1$ and grows monotonically in magnitude, from 0.16 at $r = 0$ to 2.5 at $r = 1$. The ratio of the slopes at the two ends is $1/(1 - p_0)^2 = 16$. This is the origin of the observed acceleration of the decline at high penetration. Equation (49) is an analytic (rational) function of r and contains no threshold: the suppression is a smooth crossover, whose curvature is controlled by p_0 alone. The measured steepest slopes, 1.6–2.2 at $r \geq 0.925$, are consistent with the value 2.5 at $r = 1$.

The measured W lies below W_{id} at small r (figure 3b), because the metastable upper branch decays before ρ_2 is reached; for $r \leq 0.45$ the maximum separation occurs at $\rho = 0.15$ instead of 0.16. If the measured upper branch is combined with the estimate (48) for the lower branch, the measured loop widths are reproduced to within 0.0065 for $L = 1000$ and 0.0042 for $L = 8000$ (rms deviation about 0.002 at every ring size). The remaining discrepancy between W and W_{id} is therefore entirely due to the finite lifetime of the upper branch, and the r -dependence is carried by the lower branch through $c(r)$.

6.6 Composition and fraction of standing vehicles

The fraction of vehicles that are inside the jam is

$$f_s = \frac{\phi}{\rho} = \frac{\rho - \rho_{\text{out}}}{\rho(1 - \rho_{\text{out}})}. \quad (51)$$

At $\rho = 0.15$ (jammed start, $L = 2000$), the measured fraction of vehicles with $v = 0$ follows (51) closely up to $r = 0.8$ (table 4). At $r = 0.9$, $\rho = 0.15$ lies just above $\rho_{\text{out}} = 0.134$ and the estimate (0.125) exceeds the measurement (0.088). The measured AV share among the standing vehicles equals r to within 5×10^{-4} , and the mean standing times of AVs and HVs are the same. This is expected: inside a compact jam a vehicle is held by the gap constraint, which is common to both types, and the type matters only at the head of the jam. It is also the assumption that enters (44).

Table 4: Jam-front departure rate c , outflow flux and density from (44) and (46), and the fraction f_s of standing vehicles at $\rho = 0.15$: estimate (51) against measurement ($L = 2000$, 10 seeds).

r	c	J_{out}	ρ_{out}	f_s (estimate)	f_s (measured)
0.0	0.250	0.238	0.048	0.716	0.715
0.1	0.270	0.256	0.051	0.693	0.688
0.2	0.294	0.278	0.056	0.666	0.665
0.3	0.323	0.303	0.061	0.633	0.631
0.4	0.357	0.333	0.067	0.594	0.595
0.5	0.400	0.370	0.074	0.545	0.539
0.6	0.455	0.417	0.084	0.483	0.483
0.7	0.526	0.476	0.096	0.402	0.398
0.8	0.625	0.555	0.111	0.289	0.290
0.9	0.769	0.666	0.134	0.125	0.088
1.0	1.000	0.833	0.167	0	0

6.7 Remarks

The jam-outflow picture neglects new jams nucleated by the rare braking of moving HVs, correlations between successive head vehicles, and the presence of several jams. The agreement in table 4 and figure 3 indicates that these effects are small for $p = 1/64$. The picture also suggests how the AV behavior rule enters the hysteresis: through (48), the jammed branch depends on the rule only via the mean time an AV needs to leave the head of a jam. An AV with the naive rule leaves in one step; an IDM-governed AV with $a = 0.097 \text{ cell/step}^2$ needs several steps before its rounded displacement becomes nonzero, which would lower c . This is consistent with the pilot runs in which human-calibrated IDM AVs did not raise the jammed branch, but it has not been worked out quantitatively.

7 Summary

In this note, we have derived the main analytical results behind the study of mixed AV/HV traffic in the NaSch model. We have shown that the shared safety floor $v \leq d - 1$ is equivalent to a minimum time headway of one step, so that CACC ($h = 0.6 \text{ s}$) is always clipped and reduces to the naive deterministic rule, while ACC ($h = 1.1 \text{ s}$) is not. We have rederived the exact $v_{\text{max}} = 1$

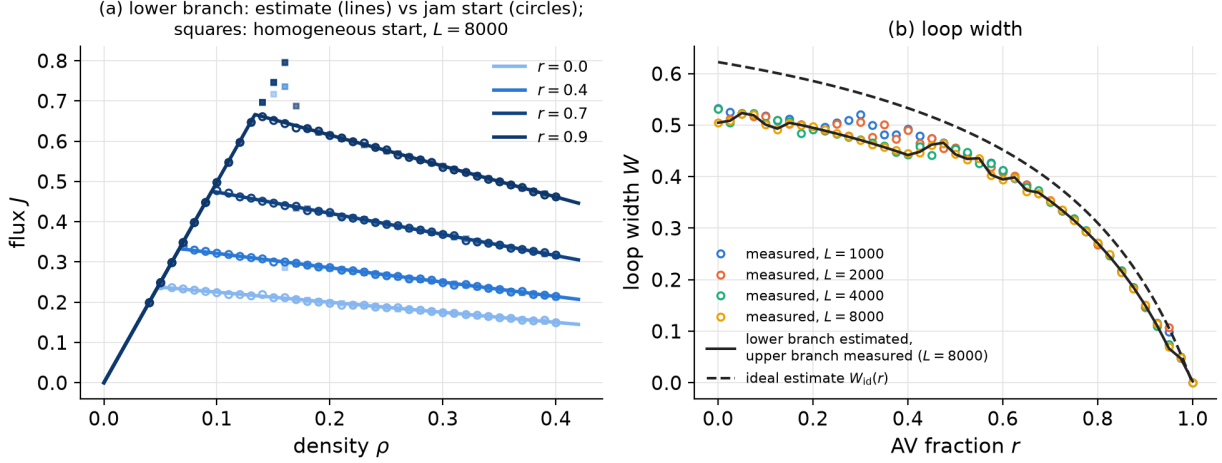


Figure 3: (a) Lower branch of the VDR model: estimate (48) (lines) against jammed-start simulations (circles); squares are homogeneous-start runs ($L = 8000$, 20 seeds). (b) Loop width against r at four ring sizes, the ideal estimate (49) (dashed), and the estimate obtained from the measured upper branch and (48) (solid).

solution [Evans, 1997] in COMF form and obtained the two-species flux in closed form, (37); it agrees with the simulation to within 0.53%, and it predicts a discontinuity of the capacity at $r = 1$ owing to the bound $g < q_{HV}$. Finally, we have shown that a jam-outflow argument gives the lower branch of the VDR model as $J_{low} = c(r)(1 - \rho)$ with $c(r) = q_0/(1 - p_0 r)$, which reproduces the simulations without fitted parameters, and which yields the loop width (49) as a smooth rational function of r . We conclude that the absence of a threshold in the hysteresis suppression, and the acceleration of the decline at high penetration, both follow from the departure statistics at the jam head.

Several points remain open: the gap between the equilibrium and simulated capacities of ACC and IDM, a quantitative treatment of the departure rate for IDM/ACC AVs in the VDR model, an approximate theory for $v_{max} = 5$, and the effect of lane changing. These are left as future work.

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